# S²-MAS

## Scaling persona-driven LLM simulation

A society-scale multi-agent simulation framework with 500 million household persona agents

**500M**
synthetic household personas

**2,925,263**
largest finance set in paper comparison

**12M**
distinct model-driven household agents

**14 GPUs**
3 h 59 min end to end

**12M households → 24M paired-condition choices → 288M monthly ledger states**

8 H100 80GB + 6 A100 40GB

### Data
Real-survey data
500M personas · 36 coded fields

### sharp decision models
SFT + RLCD · decision heads · LoRA
Full state + complete alternatives

### Genese engine
World state ↔ Brain policy intents
Batched arrays · state–event commit

### Financial experiments
8 reusable task families
Consumption · cash · repayment

## Consumer financial choices

| Model | Accuracy (%) | Q-macro F1 (%) |
|---|---|---|
| **sharp-421m-mlx** | **74.55** | **38.37** |
| sharp-4b | 74.14 | 36.73 |
| sharp-9b | 74.09 | 36.66 |
| GLM-5.3-Flash | 68.88 | 38.00 |
| Jev 1.13 | 65.26 | 37.48 |

240,458 shared test records · 119 question versions

421m MLX FP16; 4b/9b BF16; API baselines

## Six approaches, one sharp family

**sharp-421m**
421M · English · full-data SFT + RLCD

**sharp-322m**
322M · bilingual · PyTorch / MLX

**sharp-4b**
4B · head-only / LoRA + head

**sharp-9b**
9B · LoRA + decision head

**6 post-training recipes · 10 trained variants**

SFT · SFT→CE · SFT+RLCD · 4B head

4B LoRA+head · 9B LoRA+head

## sharp-322m-mlx deployment

**36,955** shared input tokens/s

**22.72× Jev API · 2.42× same-model PyTorch**

| Runtime | Latency (ms) | Input tokens/s |
|---|---|---|
| 322m · MLX | 9.10 | 36,955 |
| 322m · PyTorch | 22.06 | 15,247 |
| Jev 1.13 API | 206.77 | 1,627 |

Local FP16: one A100 40GB · batch/concurrency 1

3 Chinese cases · 60 timed calls/arm · API round trip

### 12M household application

Largest consumption gains: decision makers aged 25–34.

A higher purchase goal and tied limit add CNY 2,887 consumption per household; closing cash falls CNY 929.